Cell Biology

Dr. Reshna K. Albarzanji
B.Sc. M.Sc. PhD.
Cytogenetics
Hawler Medical University
College of Medicine
Microbiology dep.
reshna.kamal@hmu.edu.krd

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Physical properties of protoplasm

Protoplasm

Protoplasm is the living part of the cell, which comprises of different cellular organelles. It is a jelly-like, colorless, transparent and viscous living substances present within the cell wall.

The term protoplasm was proposed in the year 1835 J.E. Purkinje coined the term protoplasm and is known as the primary substance, as it is responsible for various living processes.

Protoplasm consists of many products like carbohydrates, lipids, proteins and nucleic acid.

Protoplasm mainly consists of three elements : **Plasma membrane, nucleus and Cytoplasm**

Protoplasm helps in performing various functions like transport of food materials , excretion of waste materials, growth and development ,reproduction.

In the present time, the term protoplasm is not so much used , cytoplasm is used in replace of protoplasm.

Protoplasm Functions

Protoplasm help the cell to perform all the essential steps. Various functions performed by the protoplasm includes:

*Metabolism: Protoplasm is responsible for performing metabolisms activities which helps the cell to keep alive.

*Excretion: It is very important for a cell to excrete out the waste products as these may be harmful for the cell and excretory products include urine, sweat, hormones and digestive enzymes

*Reproduction: As we know that cells divide to form identical cells and it can help in the process of reproduction of cells.

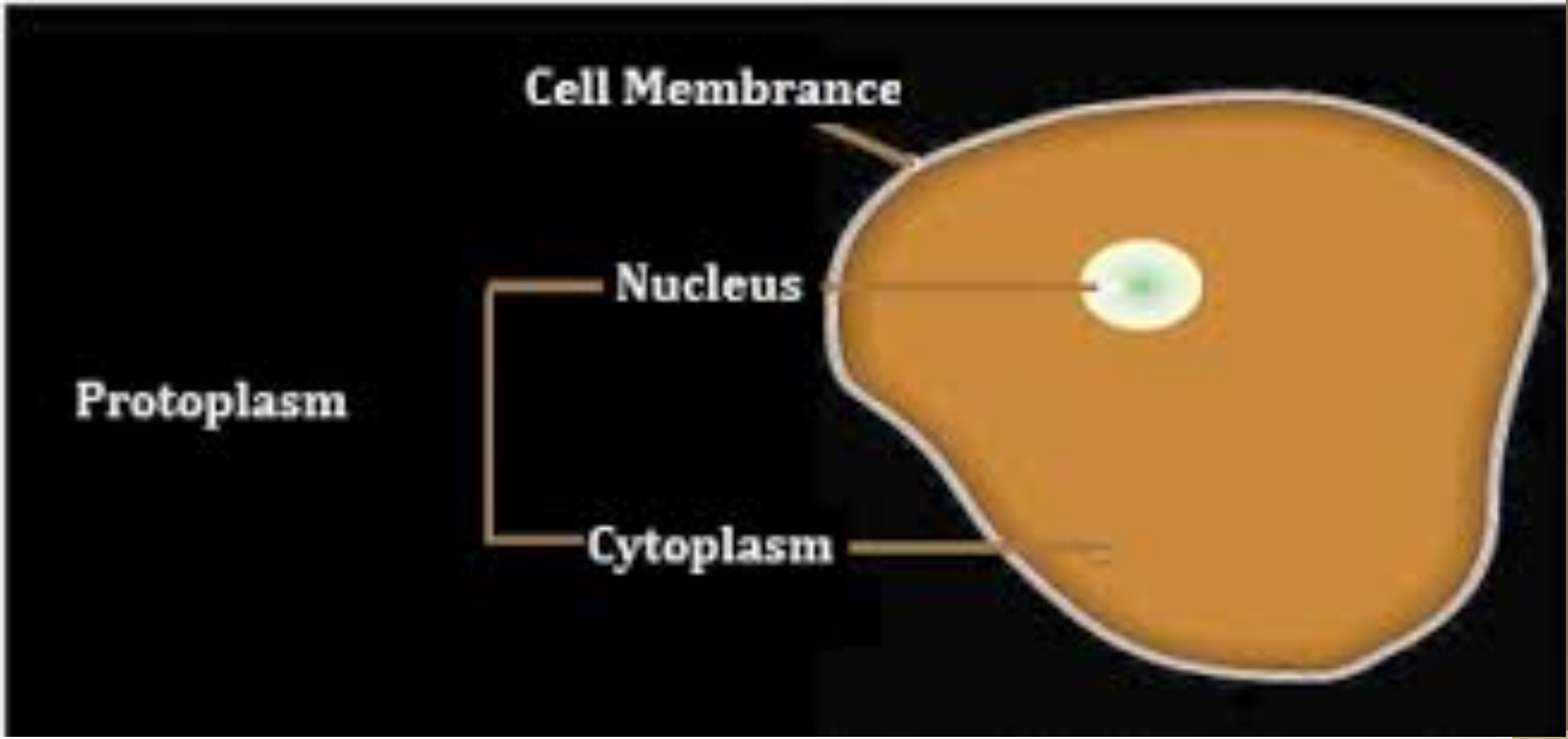
*Chemical reactions: As we have seen that cell perform many of its activities within the cell itself, and some form of energy is compulsorily required for various organisms to perform chemical reactions and the energy present in plants comes from the photosynthesis process while in the energy that comes in animal cell comes through the process of respiration.

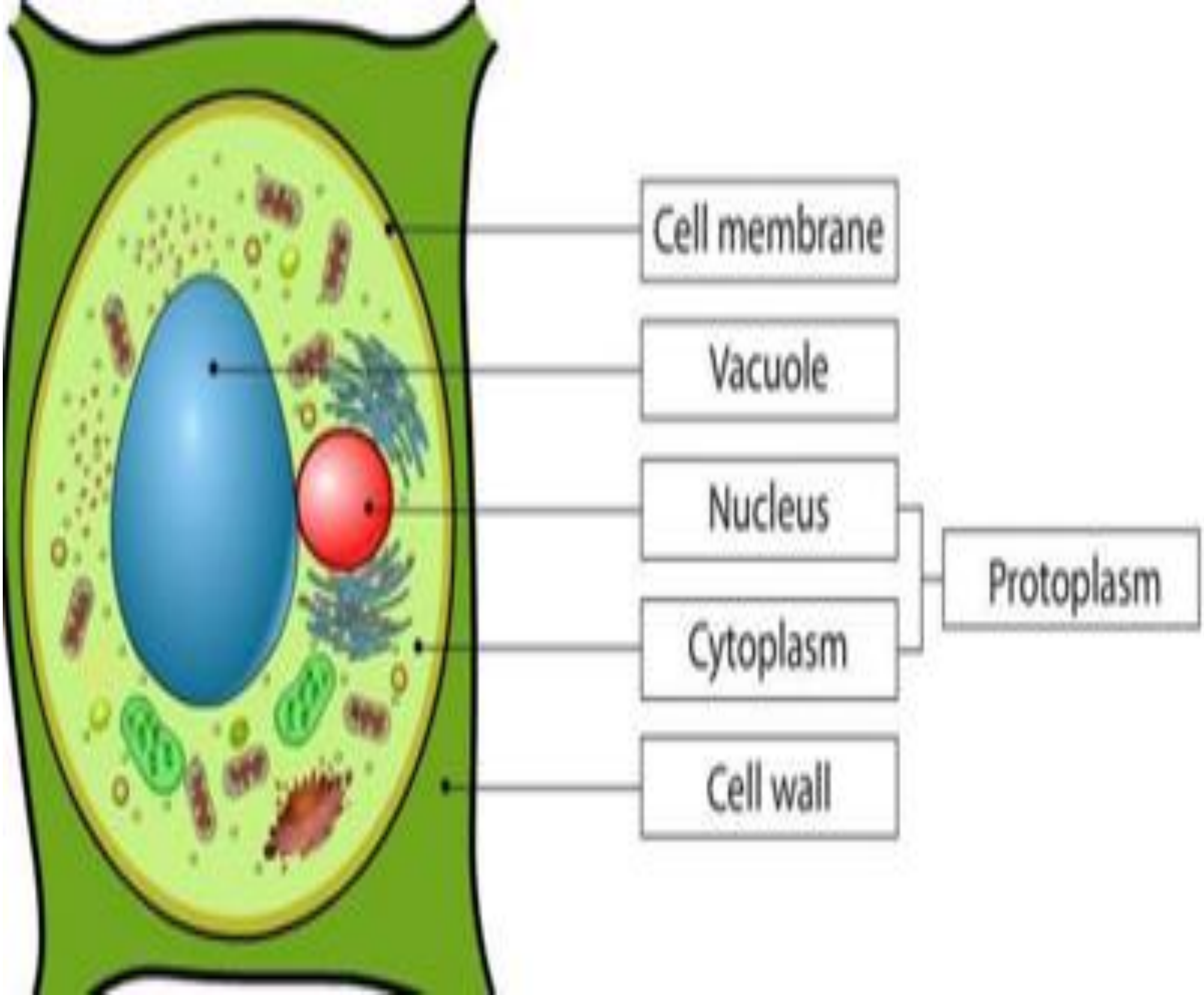
*Structure and Support: Protoplasm plays an important function in providing support and structure to the cells.

*Irritability: Protoplasm is having the ability to respond to things i.e. stimuli, and an important aspect of life of the cell.

*Transport: Cells have to transport the food and remove the waste excreted material and cells produce the nutrients used.

*Growth: increase in cell size will result in an increase in cytoplasm , and protoplasm help in growth of cells and increase in height of the cell and number of cells will also be increased.


Cell Membrance
Nucleus
Protoplasm
Cytoplasm


Cell membrane
Vacuole
Nucleus
Cytoplasm
Cell wall
Protoplasm

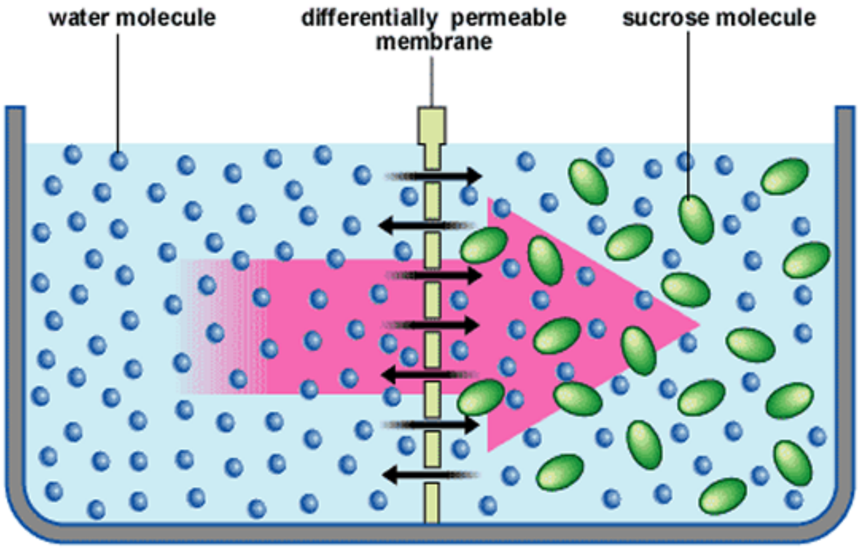
Protoplasm

- The living matter of which each cell of living organisms is built up.
- It is a jelly like substance that differ in its stiffness and viscosity due to factors such as
- Acidity
- Ionic tension
- Temperature.

Physical properties of Protoplasm

- **1.Tyndall Effect** : As protoplasm is colloidal in nature it exhibits Tyndall effects(*light scattering by particles in a colloid)*
- **2. Elasticity and Contractility** : It is due to folding and unfolding nature of proteins.
- **3. pH** : Protoplasmic pH is around 6-8.
- **4.Cyclosis**: is a cytoplasmic streaming, by which the cytoplasmic fluid inside a given cell is moved around in currents, carrying nutrients, proteins, and organelles through the cell, and it needs high energy molecules(ATP).
- **5-Permeability**: Is the traffic of different molecules across the plasma membrane
- **6-Antagonism**: the ability of protoplasm to remove the toxic action of different substances.


Solution:
Suspension:
Colloid:


water molecule
differentially permeable membrane
sucrose molecule

Chemistry of the cell

Protoplasm is a dynamic system, changes continuously, it has a specific mode in each cell.

In which is differ mainly in protein, nucleic acids and other chemical compounds from cell to cell and tissue to another even in the same individual due to

*Age of the cell

*Cell specialization

*Cell function.

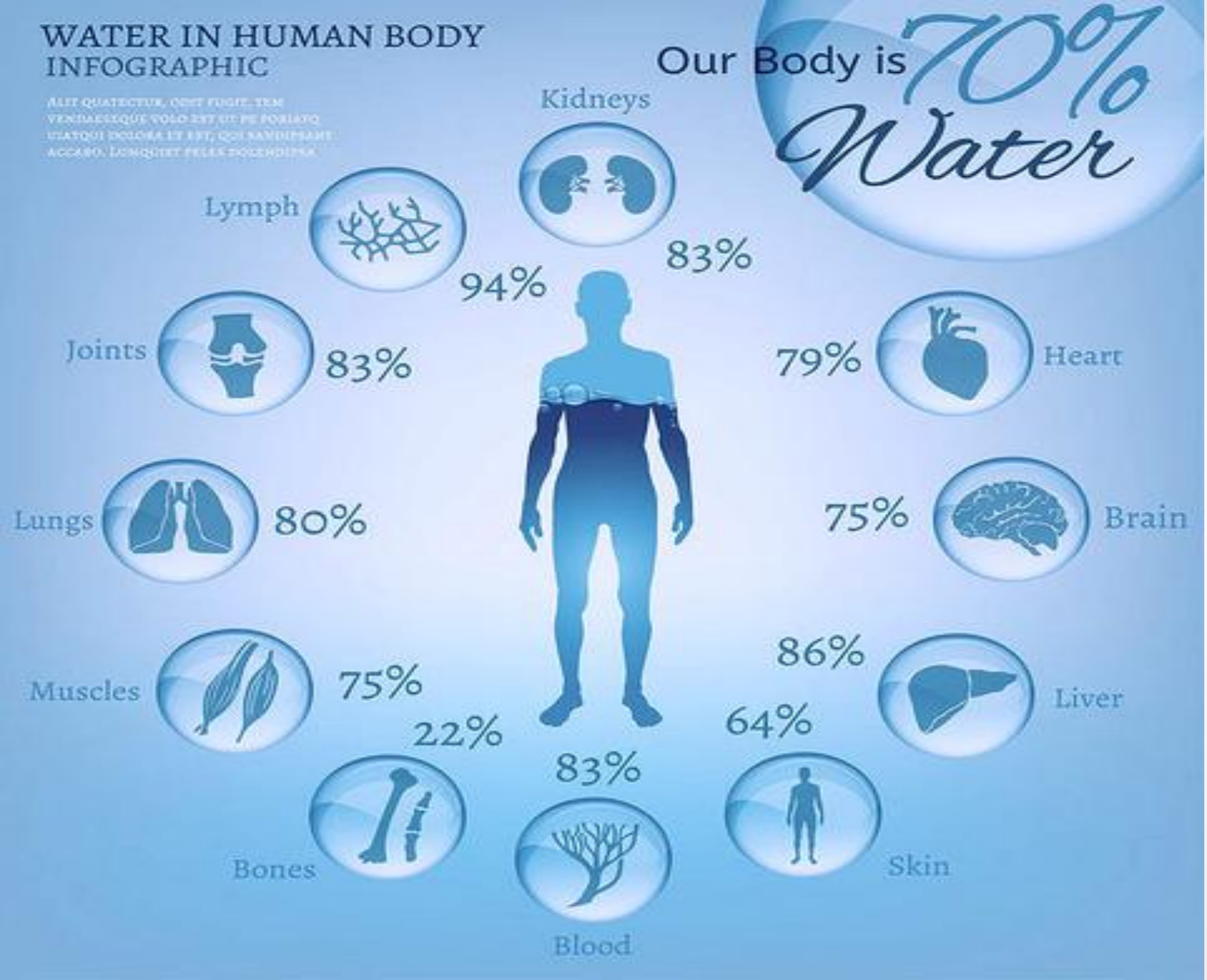
Chemical components of protoplasm

- Chemical components of protoplasm can classified to:
- 1-Water: Accounting about (70-90)% of total cell mass.
- 2-Inorganic substance: (minerals, ions)
- 3-Organic substances: (protein, carbohydrate, nucleic acid and lipid)

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

Water:

Water is an absolutely essential constituent of living matter. Water is the most abundant substance in the cell, its accounting about (70-90)% of total cell mass and this abundance is due partly to its unique chemical properties created by the influence of its hydrogen bonds these properties include the following:


WATER IN HUMAN BODY
INFOGRAPHIC
Our Body is 70% Water
Kidneys
83%
Lymph
94%
Joints
83%
Heart
79%
Lungs
80%
Brain
75%
Muscles
75%
Liver
86%
22%
Bones
83%
Blood
64%
Skin

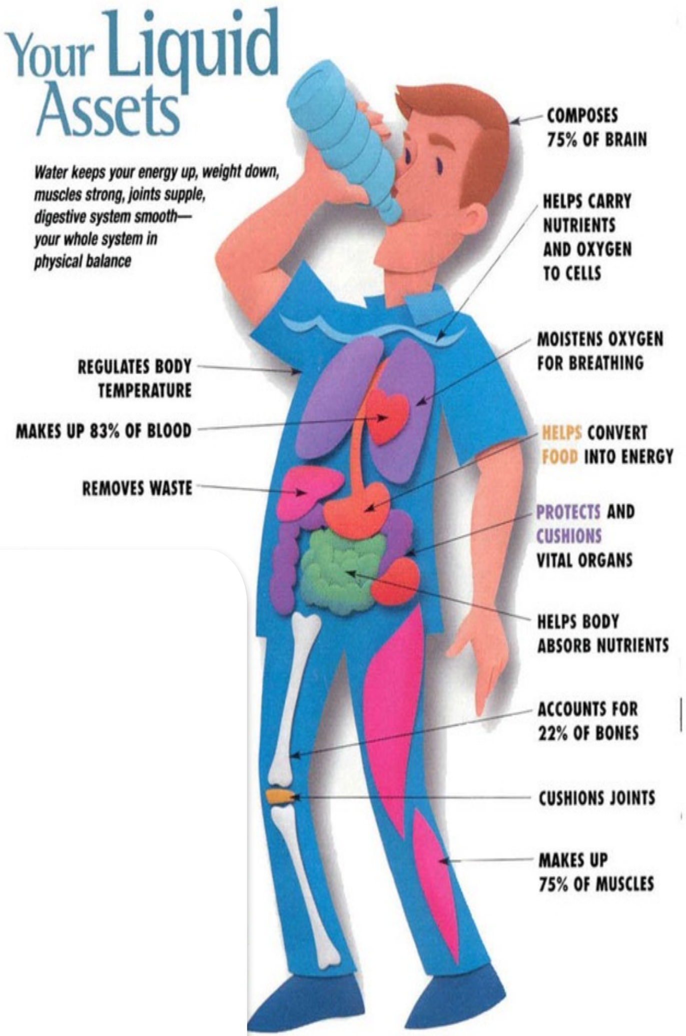
- Properties of water

1- Water is an excellent solvent

2- Water has strong surface tension.

3- Water has very stable temperature.

4- Cohesion and adhesion


Your Liquid Assets
Water keeps your energy up, weight down, muscles strong, joints supple, digestive system smooth—your whole system in physical balance
COMPOSES 75% OF BRAIN
HELPS CARRY NUTRIENTS AND OXYGEN TO CELLS
MOISTENS OXYGEN FOR BREATHING
REGULATES BODY TEMPERATURE
MAKES UP 83% OF BLOOD
REMOVES WASTE
HELPS CONVERT FOOD INTO ENERGY
PROTECTS AND CUSHIONS VITAL ORGANS
HELPS BODY ABSORB NUTRIENTS
ACCOUNTS FOR 22% OF BONES
CUSHIONS JOINTS
MAKES UP 75% OF MUSCLES

1-Water is an excellent solvent.

Ionic and polar covalent substances are soluble in water. These substances are called Hydrophilic (water loving).

Non polar covalent substances do not dissolve in water and called hydrophobic (water fearing).

Water is a good solvent

The Properties of Water

- Water can dissolve salts (ionically bonded compounds) and **hydrophilic** (water–loving) molecules because both the water and these molecules are **polar**.

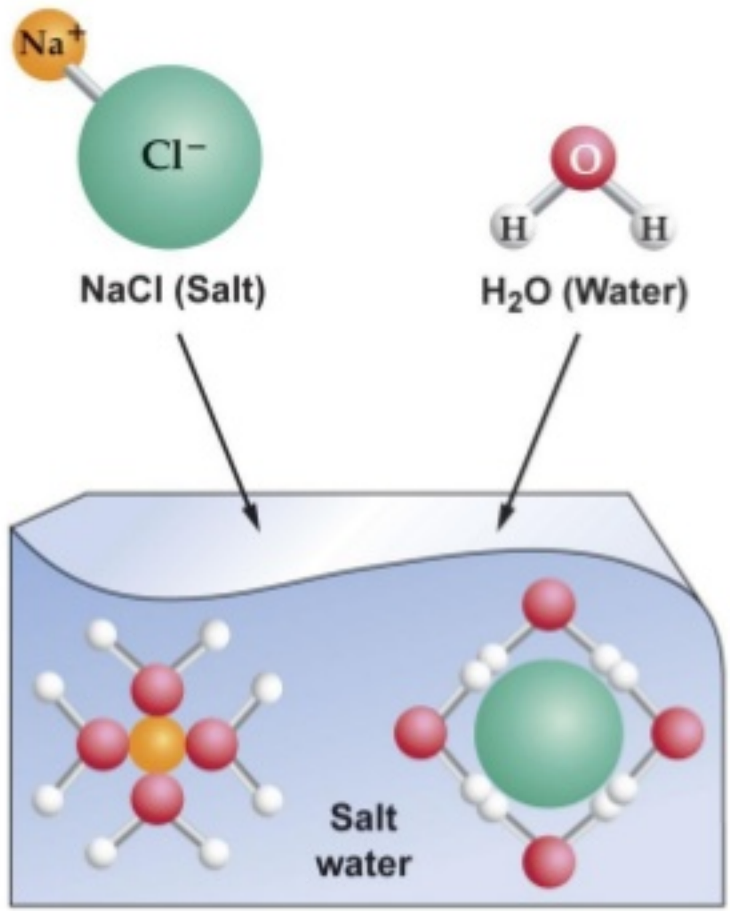


Figure 2.6

2-Water has strong surface tension.

water molecules are held together by hydrogen bonds, that gave them high degree of cohesion, or ability to stick together. As a result, water has strong surface tension.

2-Water has strong surface tension.

Surface Tension: "The property of the surface of a liquid that allows it to resist an external force, due to the cohesive nature of its molecules.

The cohesive forces between liquid molecules are responsible for the phenomenon known as surface tension.

water molecules are held together by hydrogen bonds, that gave them high degree of cohesion, or ability to stick together. As a result, water has strong surface tension.

Paper clip can float on water, due to high surface tension of water

3- Water has very stable temperature

Hydrogen bonding gives water a high specific heat.

Heat is absorbed when hydrogen bonds are break and is released when Hydrogen bonds form.

4-Cohesion and adhesion

- Water molecules stay close to each other by cohesion, due to the collective action of hydrogen bonds between water molecules.
- Water also has high adhesion properties because of its polar nature.
- Because water has strong cohesive and adhesive forces, it exhibits capillary action. Strong cohesion from hydrogen bonding and adhesion allows trees to transport water more than 100 m upward.


THANK
YOU

- What type of substance is protoplasm?
- What is the living substance of protoplasm?
- What’s the most abundant substance in protoplasm?
- Is protoplasm present in animal cell?
- Where protoplasm is found?
- What is the main physical properties of protoplasm?
- What is the important of cyclosis?
- What is the chemical composition of protoplasm short answer?
- Why water is important in our life?